

CSE 4512

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Lab 02 Task

Your company, Iutians Assemble, is expanding its headquarters and requires a new network design to support various departments. The IA headquarters consists of the following departments across multiple floors: Engineering - Floor 1, Marketing - Floor 2, Sales - Floor 3, Administration - Floor 4, Management - Floor 5

Suppose they all share the prefix 192.(i).(i+3 % 14). Below is the required devices for each of the departments:

$E(i) = (7i + 3) \bmod 41 + 10$ (Engineering)

$M(i) = (5i + 11) \bmod 26 + 5$ (Marketing)

$S(i) = (11i + 7) \bmod 36 + 5$ (Sales)

$A(i) = (3i + 17) \bmod 18 + 3$ (Administration)

$G(i) = (13i + 1) \bmod 9 + 2$ (Management)

where i is the last two digits your student ID

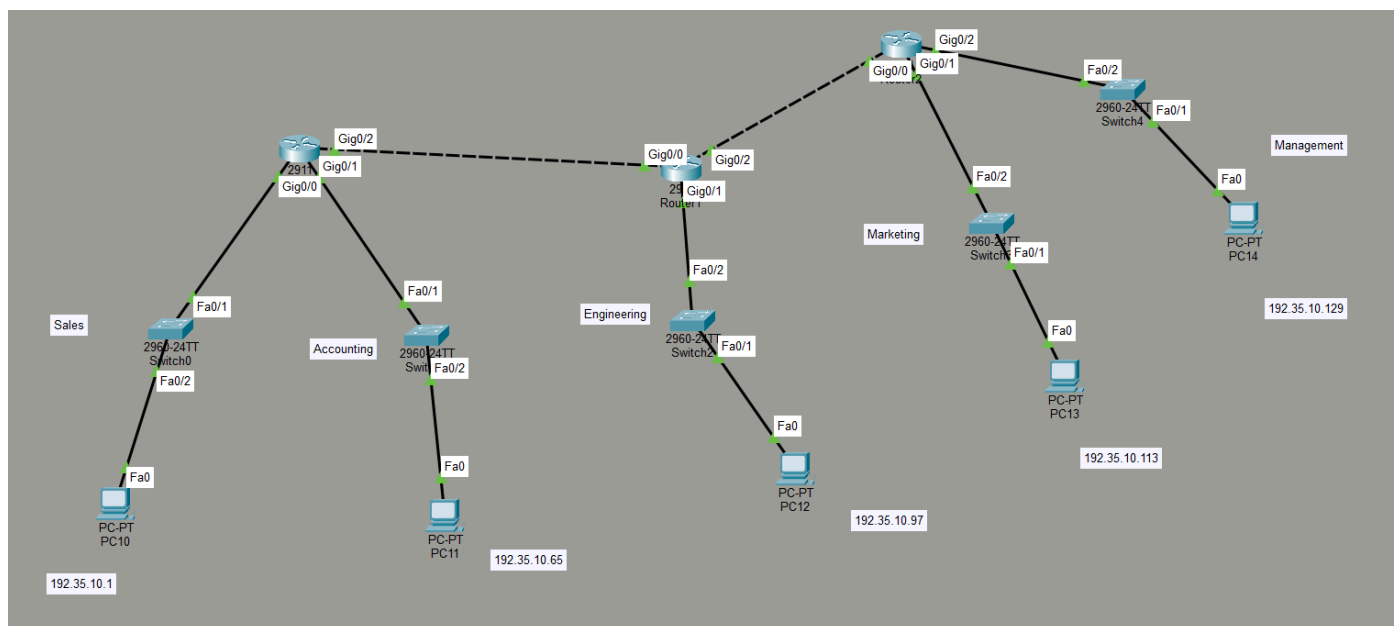
Now, by completing the following table with the most efficient design decision, design the topology with correct network flow, for each network attach at least 1 end devices.

Subnetting Worksheet:

Department	Devices	Network	Mask	Usable Host Range
Engineering	$E(i)$			
Marketing	$M(i)$			
Sales	$S(i)$			

Administration	A(i)			
Management	G(i)			

Final Topology



Procedure

1. Firstly as per the instructions, I subdivided the whole networks into subnets to better utilize the address.
2. After arranging the dept in descending order based on the host devices, I used the number of host devices in the hypothesis to find out the host bits and then the masking.
3. With proper masking I connected every end device with proper usable range and gave them their default gateway based on the ethernet they are connected onto.
4. For interconnection between routers I used /24 and different network addresses for convenience between router 1 and router2 & router2 and router3.

5. I then routed every network connected to it by using version 2 and no auto summary in every router.
6. After routing we were able to successfully send messages to every pcs connected to each other.

Observations & Results

- Ping Checking for PCs
 - Here in the following ss, I chose PC12 to to send packets and we can see that it was able to deliver packets successfully to all other PCs in the other depts. From this observation, we can come to conclusion that the connection is up and running.

```
co Packet Tracer - D:\Course\5thSem\Computer_Networks_Lab\Lab_2\220041135_lab2.pkt
PC12
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.35.10.2

Pinging 192.35.10.2 with 32 bytes of data:

Reply from 192.35.10.2: bytes=32 time<1ms TTL=126
Reply from 192.35.10.2: bytes=32 time<1ms TTL=126
Reply from 192.35.10.2: bytes=32 time<1ms TTL=126
Reply from 192.35.10.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.35.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.35.10.66

Pinging 192.35.10.66 with 32 bytes of data:

Reply from 192.35.10.66: bytes=32 time<1ms TTL=126
Reply from 192.35.10.66: bytes=32 time=1ms TTL=126
Reply from 192.35.10.66: bytes=32 time=1ms TTL=126
Reply from 192.35.10.66: bytes=32 time=2ms TTL=126

Ping statistics for 192.35.10.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 192.35.10.114

Pinging 192.35.10.114 with 32 bytes of data:

Reply from 192.35.10.114: bytes=32 time<1ms TTL=126
Reply from 192.35.10.114: bytes=32 time<1ms TTL=126
Reply from 192.35.10.114: bytes=32 time=1ms TTL=126
Reply from 192.35.10.114: bytes=32 time<1ms TTL=126

Ping statistics for 192.35.10.114:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.35.10.130

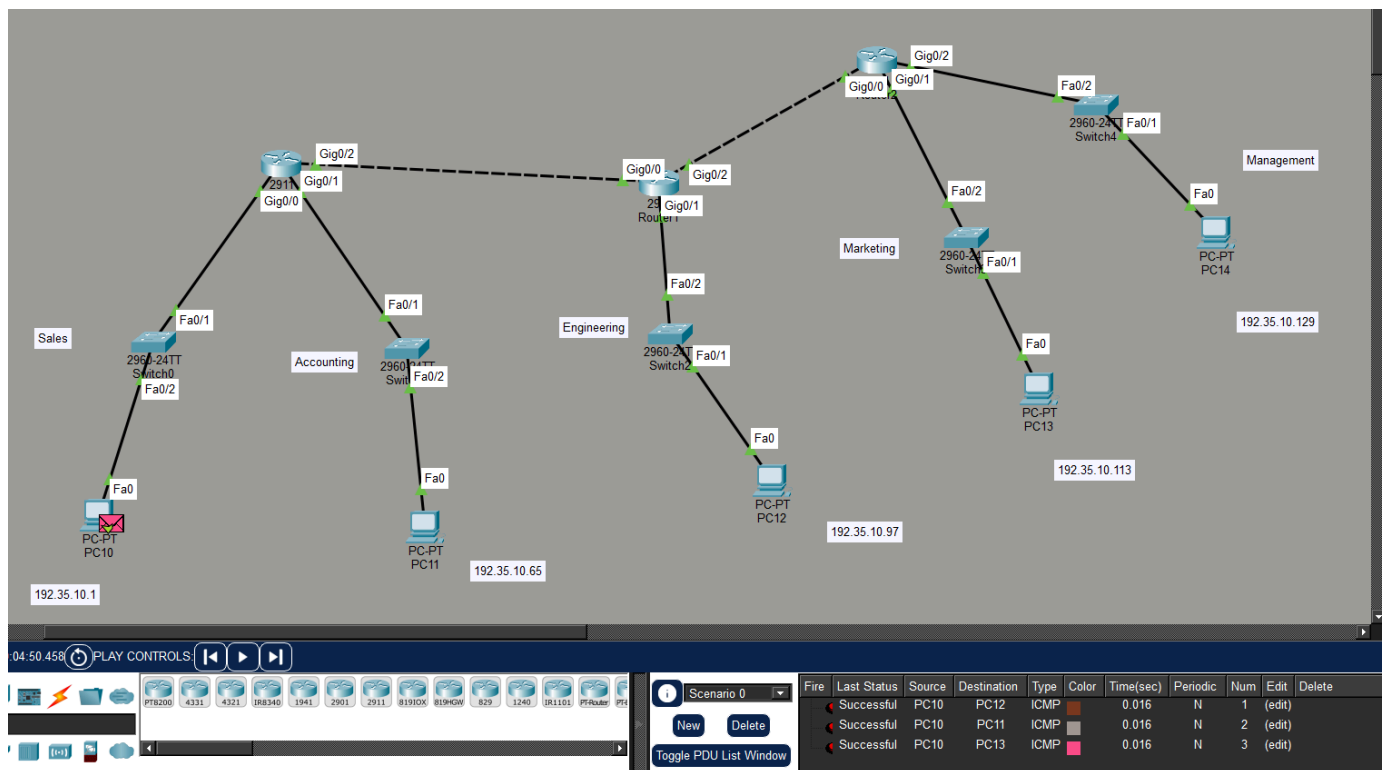
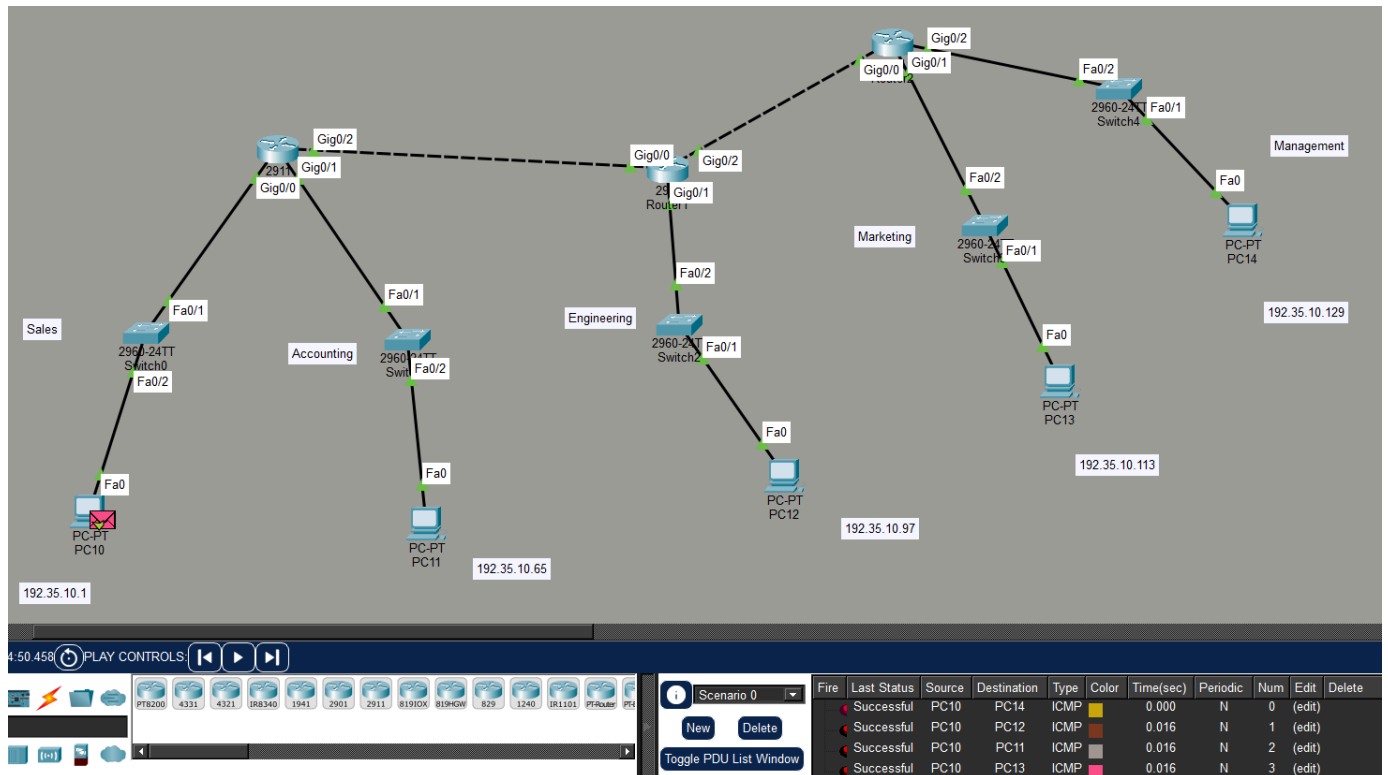
Pinging 192.35.10.130 with 32 bytes of data:

Reply from 192.35.10.130: bytes=32 time<1ms TTL=126
Reply from 192.35.10.130: bytes=32 time=1ms TTL=126
Reply from 192.35.10.130: bytes=32 time=1ms TTL=126
Reply from 192.35.10.130: bytes=32 time=1ms TTL=126

Ping statistics for 192.35.10.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- Sending messages
 - In the following SS in the bottom right you can clearly see messages were sent successfully from PC10 to all the other PCs



Challenges

- Calculating the subnet at first was a bit confusing as well as time consuming.